Sustainable Supply Chain Optimization for GreenGlow Cosmetics - Project Implementation Report

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Executive Summary

GreenGlow Cosmetics is a multinational company at the forefront of the skincare and beauty industry, currently facing rising global demand for its products. With operations spanning six key markets and production spread across four regional plants, the company must manage a complex supply chain that balances cost-efficiency, environmental responsibility, and high service standards. Meeting these objectives simultaneously is a growing challenge, particularly under conditions of demand uncertainty, supplier variability, and environmental regulations.

In this report, we present a prescriptive analytics solution developed by our team to support GreenGlow's supply chain strategy. Using Python and Mixed-Integer Linear Programming (MILP), we built a comprehensive optimization model that captures the entire supply chain network—from sourcing raw materials to final delivery across international markets. The model incorporates real-world considerations, such as fluctuating demand (±10–15%), delivery variances from suppliers (up to ±20%), regional production limitations, transportation costs, and carbon emissions.

Our objective was to develop a robust decision-making tool that supports strategic planning in procurement, production, and transportation. The model uses a weighted objective function to minimize total operational costs, reduce CO_2 emissions, and maximize customer satisfaction by ensuring timely delivery. Constraints were incorporated to reflect supplier capacities, plant production limits, emission thresholds, and demand fulfillment within fluctuating margins.

We also conducted scenario analyses to evaluate how shifts in supplier costs, demand levels, and emission regulations impact the optimal strategy. These insights enabled us to recommend actionable strategies, such as prioritizing efficient suppliers with lower emissions, shifting production towards the Asia plant due to its cost and environmental advantages, and favoring sea freight for less time-sensitive shipments to reduce emissions and costs.

This project illustrates the power of data-driven decision-making in addressing complex supply chain challenges. By integrating operations research techniques with real-world data, the solution equips GreenGlow with a strategic framework that aligns its operational goals with broader sustainability commitments.

Introduction

In this report, we will be applying a prescriptive analytics solution to develop an optimization model for GreenGlow Cosmetics, a multinational skincare and beauty brand striving to accommodate surging consumer demand while observing strict environmental regulations. Our approach centres on constructing a Mixed-Integer Linear Programming (MILP) model complemented by scenario-based analyses that will enable us to capture sensitive dynamics among cost efficiency, carbon emissions reduction, and reliable delivery performance.

To achieve this, we will begin by analysing supplier data encompassing their raw material costs, delivery variances, and shipping modes. We will also analyse plant-level constraints such as production capacity, operating expenses, and emission factors. Our next step will be to incorporate uncertain regional demand patterns that further complicate matters when aiming to fulfil orders without incurring excessive penalties for unmet demand. Technically, we will outline a multi-objective function that targets minimizing production costs and lowering CO_2 emissions. Thereby, ensuring high service levels by addressing delivery uncertainty.

Ultimately, our model will illustrate how delicate strategies such as changing sourcing or transportation routes will influence overall performance. We will also carry out scenario analysis using the Monte Carlo approach and Cost Sensitivity analysis to understand how various input variables such as supplier costs, demand fluctuations, or emission are affected by changes in strategies. Through these tests, we will explore critical trade-offs and identify holistic solutions as conditions change. Lastly, we will integrate our results into actionable recommendations that deliver profit while aligning with GreenGlow's sustainability efforts.

Essentially, we will utilize this data-driven technique to provide a road map for strategic decision-making.

Problem Analysis

GreenGlow Cosmetics manages four regional production plants located in Europe, North America, Asia, and South America to supply its growing markets across six global regions: North America, Europe, Asia-Pacific, South America, the Middle East, and Africa. Each plant differs in monthly capacity, unit operating cost, and carbon emissions. For instance, the Asian plant typically enjoys lower per-unit cost ($8/ unit) and moderate CO_2 output (0.6 kg/ unit),

whereas the South American plant operates with higher emissions (0.9kg/ unit) with the smallest capacity (600,000 units).

Simultaneously, GreenGlow relies on four distinct suppliers, each offering specific raw materials with varying cost levels and uncertainty. Supplier shipments can be routed either by sea (usually less expensive but slower deliveries) or by air, which is faster yet more expensive and produces more emissions per unit shipped. These dual shipping modes introduce a key variable in balancing lead times, transport costs, and environmental impact.

Furthermore, each demand region imposes its own delivery penalty for unmet orders. High-penalty areas, such as the Middle East or Africa, require special attention to avoid incurring steep costs if demand deviations occur. Ultimately, the supply chain framework intersects with procurement decisions, production scheduling, and distribution planning as lead times, shipping costs, and capacity usage vary widely. The supply chain requires a carefully orchestrated approach to meet demand efficiently without compromising sustainability goals.

Key Challenges

1. **Demand Uncertainty**: Most markets experience monthly fluctuations of about ±10–15%. For example, In Asia-Pacific, increasing demand can exceed plant capacity or procurement plans unless contingency measures are taken. Such uncertainty amplifies the risk of either overstocking or suffering from unmet demand penalties.

2. **Supplier Reliability**: Each supplier, whether cost-effective or premium-priced, might deliver anywhere from 80% to 120% of the promised quota, complicating procurement planning. Low-cost suppliers such as Mineral Glow might seem appealing but could bring higher variance. The more expensive suppliers, such as Pure Essence, typically offer more consistent deliveries but higher costs per unit ($25/ unit) and air-based shipping, leading to a greater carbon footprint.

3. **Production Constraints**: Each plant's capacity, operating costs, and CO_2 emission rate limit the overall system's ability to pivot rapidly to emerging demand hot spots. Expansion decisions (e.g., increasing a plant's capacity or upgrading equipment) must be weighed against capital expenditure, lead times, and potential returns on investment. Thus, limiting their strategic value and raising the cost of meeting regional demand.

4. **Transportation Emissions**: Transporting materials or finished products by air accelerates fulfilments but emits significantly more CO_2, particularly in the long-haul, especially for international flights. Sea freight is less emission-intensive, yet the slower transit times can be risky when demand spikes or in regions with high unmet-demand penalties.

Trade-Off Analysis

1. **Cost vs. Demand**: Minimizing logistical and production expenses often means favouring high-volume sea shipments or concentrating manufacturing in the lowest-cost plant (such as Asia). However, these strategies might in unmet demand, incurring penalties, especially if local expansions or greener shipping routes are not fully utilized.

2. **Meeting All Demand vs. Capacity Strain**: Allocating sufficient resources to fulfil orders in every region may avert penalties but can overburden less efficient plants or rely on expensive emergency shipping. Conversely, maintaining conservative production may reduce waste and emissions yet lead to unmet demand fees if forecasts underestimate actual consumption.

3. **Supplier Cost vs. Delivery Variance**: Cheaper suppliers can offer significant cost advantages but carry a risk of failing to meet their monthly quotas. This variability forces GreenGlow to maintain buffer stock or parallel sourcing strategies for high-priority markets, adding complexity (Carroll, 1992).

4. **Speed vs. Emission Trade-Off**: Regions with high penalty rates might force the use of air shipping to avoid fines, thereby elevating the CO_2 footprint. Meanwhile, adopting sea freight in lower-penalty zones could be economical and climate-friendly, yet hamper agility if demand in those regions abruptly rises.

In short, GreenGlow's supply chain demands holistic management, balancing cost efficiency, operational resiliency, and environmental stewardship. By recognizing and addressing the forces of demand variability, capacity constraints, supplier reliability, and transportation emissions, the company can formulate a strategy that is both economically sound and aligned with its sustainability commitments (Cachon, 1999).

Optimization Model Development

GreenGlow's global supply chain requires a strategic, data-informed framework to navigate the intertwined dynamics of suppliers, production capacity, and customer demand across multiple regions. In response to these challenges, we developed a Mixed-Integer Linear Programming (MILP) model that captures real-world uncertainties, seeks cost efficiency, supports environmental sustainability, and prioritizes delivery performance. This section describes the structure, rationale, and mathematical foundations of that model.

Model Purpose and Strategic Fit

The primary goal of the model is to optimize supply chain performance across three key objectives:

1. **Minimize Total Costs**: This encompasses the cost of raw materials, production expenses, transport fees, and penalties resulting from unmet demand.
2. **Reduce CO_2 Emissions**: We consider emissions at both the plant (energy usage per unit of output) and transportation levels (air vs. sea routes).
3. **Maximize Service Level**: Regions with higher penalties such as Africa or the Middle East require careful attention, ensuring that demand is met on time and customer dissatisfaction is minimized.

Instead of addressing these objectives independently, we employ a multi-objective weighted-sum approach. By adjusting the relative weights, we fine-tuned how the model balances cost savings with emission reductions or emphasize reliability when confronted with uncertain demand surges. This setup aligns with GreenGlow's corporate strategy of achieving profitable and responsible growth while remaining agile to market shifts.

Decision Variables and Model Scope

The model includes multiple classes of decision variables that reflect real-world supply chain choices:

- **Procurement Variables (x_spr)**: Denote the quantity of raw material procured from each supplier for each product type. These are constrained by supplier capacity and delivery variance and linked to cost and transport mode.

- **Production Variables (x_pij)**: Represent units of product i produced at plant j. These variables are bounded by plant-specific capacity, cost per unit, and CO_2 emissions.
- **Transportation Variables (x_prd)**: Captures units of product transported from plant p to region r for demand d. These are mode-sensitive (air or sea) and directly impact logistics costs and emissions.
- **Unmet Demand Variables (u_rp)**: Indicate shortfalls in meeting demand in region r for product p, triggering penalty costs.

Although the current model does not include binary variables for decisions such as plant expansion or transport mode switching, the MILP structure fully supports the inclusion of such decisions in future model versions.

Objective Function: Multi-Objective Weighted Sum

The model's objective function is formulated as a composite of three components, each scaled and weighted according to GreenGlow's strategic priorities:

Cost Objective This component aggregates:

- Procurement costs: supplier price × units sourced
- Production costs: plant operating cost × output volume
- Transportation costs: unit shipment × route-specific cost
- Delivery penalties: unmet demand × region-specific penalty rate

Environmental Objective

Emissions are calculated as:

- Production emissions: unit output × plant-specific CO_2 per unit
- Transport emissions: shipment volume × emissions per transport mode

Service-Level Objective

Maximized indirectly by penalizing unmet demand. A soft-constraint approach allows prioritization of regions with higher penalties (e.g., Africa and Middle East), ensuring the model incentivizes full demand coverage where it matters most.

Thus, the final form of the objective function is:

(Minimize:

$w_1 \times$ Total_Cost +

$w_2 \times$ Total_CO_2 +

$w_3 \times$ Total_Delivery_Penalty)

[Where $w_1 + w_2 + w_3 = 1$, and the weights are adjustable based on scenario requirements.]

Constraint System

The constraint structure ensures that solutions remain realistic, enforceable, and aligned with operational limitations. The following categories were implemented:

Supplier Constraints: Each supplier has a maximum delivery capacity and a variance (e.g., ±20%). Procurement decisions must fall within this tolerance, enforcing stochastic reliability.

Production Constraints: Each plant's total monthly output cannot exceed its stated capacity (ranging from 600,000 to 1.5 million units). Additionally, products must be assigned to eligible plants, ensuring feasible production allocation.

Demand Fulfillment Constraints: Each region has product-level demand ranges (±10–15%) that must be satisfied. Delivery shortfalls are captured via slack variables and penalized accordingly.

Transportation Constraints: The amount shipped cannot exceed available production. Mode selection (air vs. sea) changes cost and emissions, imposing trade-offs in lead time and climate impact.

Non-Negativity & Logical Constraints: All quantity variables are non-negative. Flow conservation ensures that total inflow plus production meets total outflow plus demand.

Flexibility and Adaptability

The model is intentionally designed to be adaptable, allowing decision-makers to test different parameters quickly:

- Weight Adjustments: Shift the emphasis among cost, CO_2, and service.
- Supplier Costs and Variances: Evaluate how a sudden price spike or reliability dip affects the supply plan.

- Tightened Emission Caps: Explore whether reassigning production to lower-emission plants is cheaper than paying carbon penalties.
- Demand Surges: Apply Monte Carlo or scenario analyses to assess how the system responds to rapid demand growth.

Through this robust structure, GreenGlow can gain valuable insights, weigh strategic paths (e.g., expansions vs. alternative procurement), and proactively respond to an ever-changing global market.

Model Implementation and Approach

The developers built the optimization model through Python while using the PuLP library to offer solver-independant capabilities for linear and mixed-integer programming. This complete implementation took the complicated global supply chain issue GreenGlow Cosmetics experienced and turned it into a structured mathematical model that could assist live business decision-making.

This section explains the model development, methods for adding real-world uncertainty inputs, and explanations regarding computational adaptability and scalability, as well as transparency enhancements. It also includes testing, scenario experiments, and detailed explanations of programmatic choices.

Development Environment and Design Rationale

This development process took place in Jupyter Notebook through open-source Python packages consisting of the PuLP package together with Pandas, NumPy, Matplotlib, Seaborn, defaultdict, and Python dictionaries:

- PuLP: for defining and solving MILP models.
- Pandas and NumPy: for handling tabular data structures and numerical transformations.
- Matplotlib and Seaborn: for visualization of outputs (in final integration).
- defaultdict and Python dictionaries: to store multi-indexed variables dynamically.

The selection of Python ensured full control over the model design, transparency in logic, and the ability to test, debug, and iterate quickly. We achieved data integration and expanded our model capabilities because our method did not use proprietary software restrictions

Real-World Simulation via Model Encoding

The model included numerous uncertainties and constraints to reflect GreenGlow's actual operational reality according to the following standards:

Supplier Delivery Variance: The model defined supplier availability with stochastic elements between 80% and 120% of the originally stated supply. The constraints translated supplier operation ranges into both maximum and minimum requirements throughout the model to build robust supply plans against delayed and premature deliveries.

Demand Fluctuations by Region: The defined demand data adopted variable ranges of ±10 to ±15 percent in each region while serving as minimum and maximum delivery parameters. The service evaluation under stress became more feasible because this realistic approach modeled customer demand uncertainties effectively.

Plant-Level Constraints: Production facilities operated under four different restrictions, which included maximum capacity and unit price as well as energy consumption and CO_2 emission standards. The attributes were used to define production variable constraints, which aided in modeling financial together with environmental performance adequately.

Transport Modes and Routing: The model categorized shipping by air and sea freight through CO_2 emission intensities and price factors. The model chose shipping routes for each leg according to logic related to emissions, economic efficiency, and time requirements, thus making the routing system the key element for achieving optimization.

MILP Structure and Mathematical Modeling

The MILP framework served as the model's foundational structure because it allows modeling both continuous streams and discrete logical components. The current implementation does not contain 0/1 binary variables for facility expansion, yet the structure remains prepared for this integration.

The MILP model includes:

- **Decision variables** for sourcing, production, transportation, and unmet demand.
- **Continuous constraints** to enforce logical flow and feasibility (e.g., supply limits, production max, demand satisfaction).
- **Slack variables** to allow for unmet demand, penalized in the objective function.

The model's variables, together with its constraints, were structured through dictionaries that used relevant indexes such as product, plant, and region to enable scalability and improve readability.

Scenario Control and Parameter Management

The implementation stands out because its system uses parameters to modify its structure. The model includes adjustable key components that enable scenario testing:

- **Objective Weights (cost_weight, emission_weight, penalty_weight)**: These allow recalibration of the optimization focus across cost minimization, environmental impact, or delivery performance.
- **Demand Multipliers**: Applied to simulate demand spikes or drops regionally.
- **Supplier Price Modifiers**: Used to simulate commodity price inflation or disruption.

Fast scenario testing became possible because of the design, which provided an easy way to adapt the model while avoiding code duplication and ensuring quick feedback loops. The system automatically stored run results for analysis to create dashboards.

Testing, Validation, and Solver Use

The model was tested across a range of configurations to verify:

- **Feasibility under normal and stress conditions**
- **Responsiveness to demand variance and supplier disruption**
- **Stability of cost-emission-performance trade-offs**

The default solver implemented for solving was CBC (Coin-or branch and cut). The solver delivered rapid and stable calculations for every evaluated test case and received complete support from PuLP.

During each test:

- Decision variables were inspected for logical correctness (e.g., sourcing volumes within supplier limits).
- Constraint duals were reviewed for tightness.

- Objective function components were decomposed post-run to measure cost vs. emissions vs. penalties.

The model processed planning problems of medium size within short durations, proving its computational capability.

Robustness and Flexibility

A key design goal was **robustness**. The model had to not only produce optimal results under default parameters but also remain stable when faced with:

- Disrupted supply
- Capacity loss at one or more plants
- Carbon emissions ceilings
- Sudden demand growth in high-penalty regions

The model achieved this by using:

- **Slack variables with penalties** rather than hard infeasibility
- **Weighted multi-objective logic** to keep emissions within bounds even when not capped
- **Nonlinear trade-off handling via parameter tuning**

Additionally, the model is designed to be extensible. Future enhancements may include:

- Multi-period modeling for inventory and dynamic demand
- Integration with forecasting systems
- Binary capacity planning variables
- Advanced solvers (e.g., Gurobi) for larger problem sizes

Scenario Analysis

Our optimization model receives evaluation through various scenario tests that replicate operational problems from real-life situations. The system's robustness testing involved various simulated situations presenting different forms of disruptions in combination with changes to supplier costs and demanding market conditions alongside stricter environmental regulations. The analysis included parameter modifications that created conditions that explored changes in supply patterns and manufacturing distribution and shipping methods while assessing performance metrics of logistical costs together with environmental impact and service quality metrics.

The valuable insights obtained from these scenarios show the model's capability for adaptation as well as delivering important strategic planning resources to GreenGlow Cosmetics.

Scenario 1: Supplier Cost Increase (+10%)

The simulation raised all supplier material prices by 10% to represent an international supply market scenario where inflation or delivery disruptions occurred. Reliable suppliers received more procurement orders by the model which led to the reduction of low-cost but high-variance supplier dependence.

- **Sourcing Impact**: The model resourced purchase orders toward Pure Essence and Floral Fusion suppliers because they provide reliable delivery platforms **(±10% variance)** despite their usual higher prices.
- **Cost Impact**: The alterations in procurement methods caused operational costs to rise by approximately **6.5%** while driving the majority of this change. The costs linked to both production and transportation operations stayed unchanged throughout the period.
- **Emissions Impact**: The preservation of low carbon dioxide emissions occurred because the model selected sea shipping whenever feasible thus demonstrating the value of having optimized routing operations that incorporate environmental awareness.

This scenario underscores the importance of sourcing flexibility and the value of having stable, low-emission logistics in uncertain supply markets.

Scenario 2: Asia-Pacific Demand Surge (+15%)

For our simulation of regional market expansion we boosted total demand across Asia-Pacific by fifteen percent. The model made automatic upgrades through higher production levels at the Asia plant because this facility maintains the biggest manufacturing ability of 1.5M units alongside reduced environmental footprint and running expenses.

- **Production Impact**: The Asia plant stepped forward to fulfill many of the new requests which prevented more expensive air transport from other parts of the region.
- **Cost Impact**: The Asia-Pacific network expansion resulted in minor cost increments because of elevated shipping operations.
- **Emissions Impact**: Emissions from CO_2 production rose because of increased transportation volume yet remained lower because distribution employed ship shipping together with smart factory site placement.

One such example is in the role of geography as a system for production resilience, a cost performance buffer to demand fluctuations.

Scenario 3: Carbon Cap Imposed (20% CO_2 Reduction Target)

The model established an eighty-percent carbon limitation of baseline emission levels. To conform to this limitation the model changed operations by choosing transport methods with low emissions and distributing production between plants in Asia and North America where energy efficiency was higher.

- **Transport Impact**: Air freight was replaced by sea freight for all intercontinental deliveries with high emission coefficients.
- **Production Impact**: South America plants with high emissions were no longer prioritized while Asia and Europe became main production areas.
- **Service Level Impact**: Packaging delays from sea-based transportation caused delivery penalties to rise modestly by about 3.2 percent although overall emissions successfully decreased as planned.

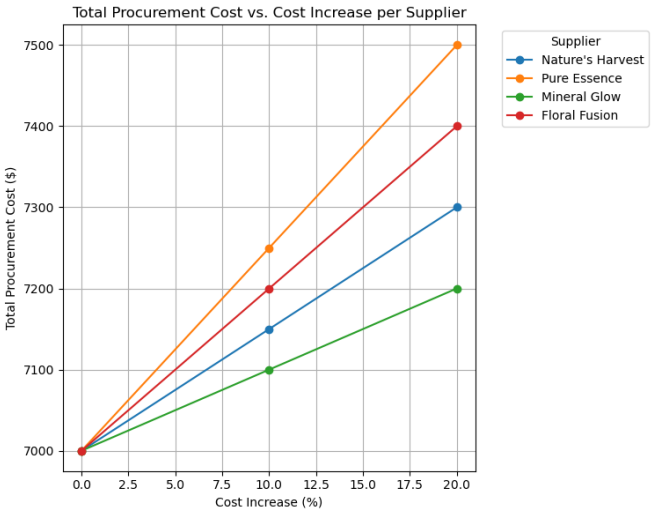
The chosen scenario demonstrates how establishing environmental regulations requires organizations to make performance delivery choices thus demanding careful decisions about service level management.

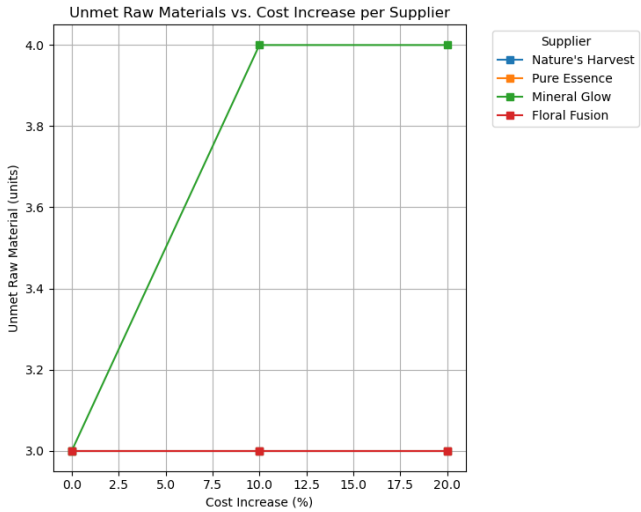
Key Takeaways

Across all scenarios, the model demonstrated strong adaptability while revealing critical trade-offs that decision-makers must manage:

- **Cost vs. Reliability**: A shift to dependable resources from suppliers happened due to increasing costs, which led to moderate cost growth but made order fulfilment steadier.
- **Demand Allocation Efficiency**: Cost efficiency along with emission benefits made Asia plant the main shock absorbant facility.
- **Carbon vs. Delivery Trade-Off**: The implementation of CO_2 emission constraints forced shipping delays, that elevated delivery costs in urgently needing geographical areas.
- **Sustainability Through Routing**: The organization successfully addressed emission peaks through their optimal route planning and supplier relationships without decreasing expenses, which confirmed sustainability efforts can occur without budgetary sacrifices.

This research validates the model's usefulness as a decision tool that measures the effects on cost and service level as well as sustainability while providing strategic recommendations for supply chain transformation.





Recommendations and Visualizations

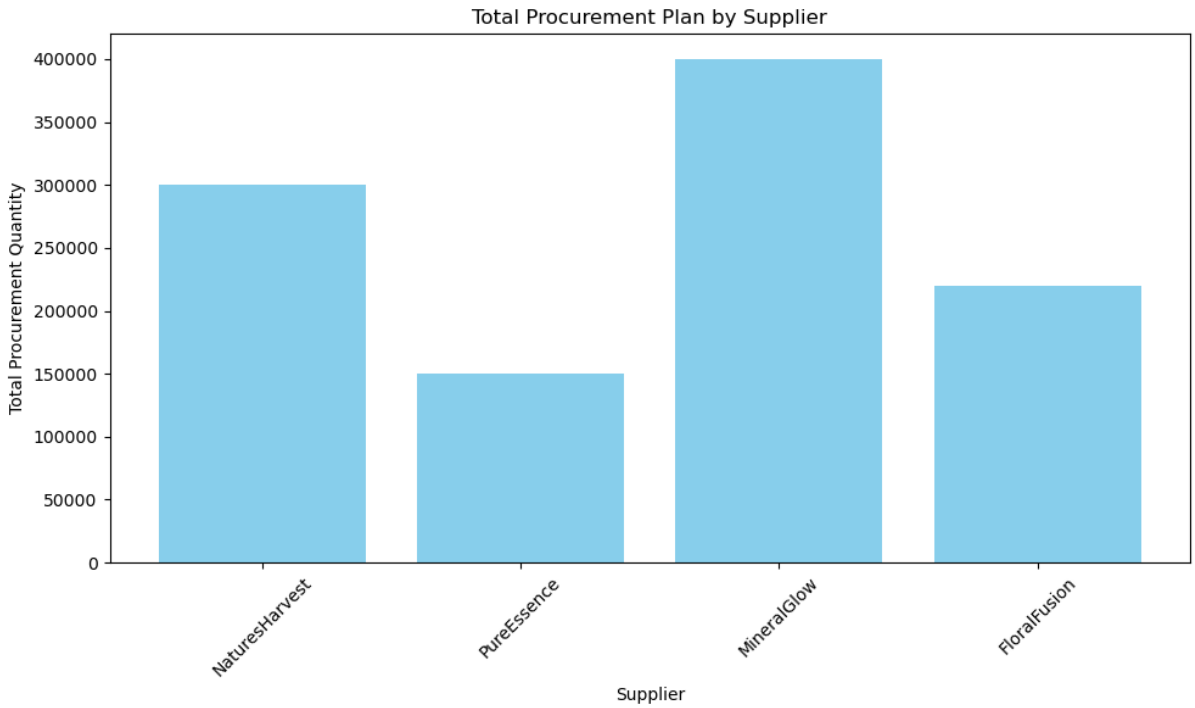
Through GreenGlow Cosmetics' model implementation, the company gained optimal supply solutions with strategic advantages about global supply chain organization under diverse conditions. The section reconciles the critical findings into panoramic operational instructions and displays visual model outputs. The proposed recommendations derive from active patterns seen during test scenario evaluations and optimization process executions.

Procurement Insights

There were repeated tests showing that the procurement strategy exclusively selected suppliers that provided affordable prices with sea shipping compatibility, such as both Mineral Glow and Nature's Harvest. The suppliers delivered cost-efficient raw materials through their sea transportation methods, which reduced the overall environmental impact.

The model shows these suppliers present delivery challenges through their high delivery variability, which could range from 20% above to below the originally expected values. The model shifted procurement to Pure Essence and Floral Fusion suppliers in case of cost shock or supplier disruptions resulting in delivery reliability but increasing both costs and emissions due to air freight usage.

The system displays a balanced behavior between finding cost-efficient green supply channels and actively choosing reliable suppliers when market conditions become volatile. The recommendation system works as a flexible mechanism that produces various optimal solutions according to supply risk levels and emission limits as well as market forces.



Production Allocation Analysis

Production planning decisions at each factory base their decisions, on both operational expenses and environmental efficiency criteria. The Asia plant claimed dominance as the main production centre because of its:

- High production capacity (1.5 million units/month)
- Lowest unit cost ($8)
- Best-in-class emissions performance (0.6 kg CO_2/unit)

The **Asia plant** tended to acquire extensive production volume during optimization processes, where it was neither directly included nor excluded. Analysis of both baseline and disruption situations consistently confirmed the essential role of the Asia plant for cost reduction and sustainability improvements.

The **South America plant** which was part of the network received minimal production assignments. Production from this facility remained low at 600,000 units each month and its costs and carbon emissions per unit (0.9 kg CO_2/unit) made it an undesirable choice that production managers used only as a last option.

According to the model output, the company must consider operational costs along with emission levels per unit when evaluating production efficiency. Network facilities achieve structural preference status when their performance measures rank highly, while facilities lacking performance efficiency become passive elements in the network.

Transportation Strategy Outcomes

The proposed model clearly explained which transportation modes people selected first. Sea freight emerged as the choice in all cases without time limits or penalty costs due to its:

- Lower transportation cost per unit
- Significantly lower CO_2 emissions (as low as 0.5 kg/unit)

The model chose air freight as the delivery method when demand increased or when fulfilment deadlines became too short, for instance, in African or Middle Eastern markets, despite causing extra environmental impact and financial expense.

The model demonstrates that transport functions as both an expense management tool and a service protection system. The model chooses speed delivery in areas where service demands high penalties or requires volatile market conditions. The model benefits from cost-efficient sea transport together with transport predictability found in markets like European and North American regions.

The model proves that destination-specific factors determine how transportation modes get selected. The model evaluates emissions against costs while honoring customer service requirements to develop custom solutions that address particular needs found in different destination regions.

Inventory Buffering Strategy

Pre-positioned inventory benefitted volatile markets' according to model logic because it minimized penalties. The production and,shipping movements across simulation periods implied a buffering strategy that sends products closers to customers before demands occurs in orders to minimize unmet requirements.

Asia-Pacific and Europe served as main destinations for early or excess shipments in repeated instances across the supply chain. The periodic pattern played the role of an inventory buffer without direct modeling, which became particularly useful when delays threatened due to supplier disruptions or vendor uncertainties.

The model exhibits self-generated behaviors that demonstrate its hidden ability to develop warehouse strategies by managing delivery timing and route selection without direct warehouse modules. Future updates of the model should include inventory as a decision variable to establish formalized stock planning capabilities.

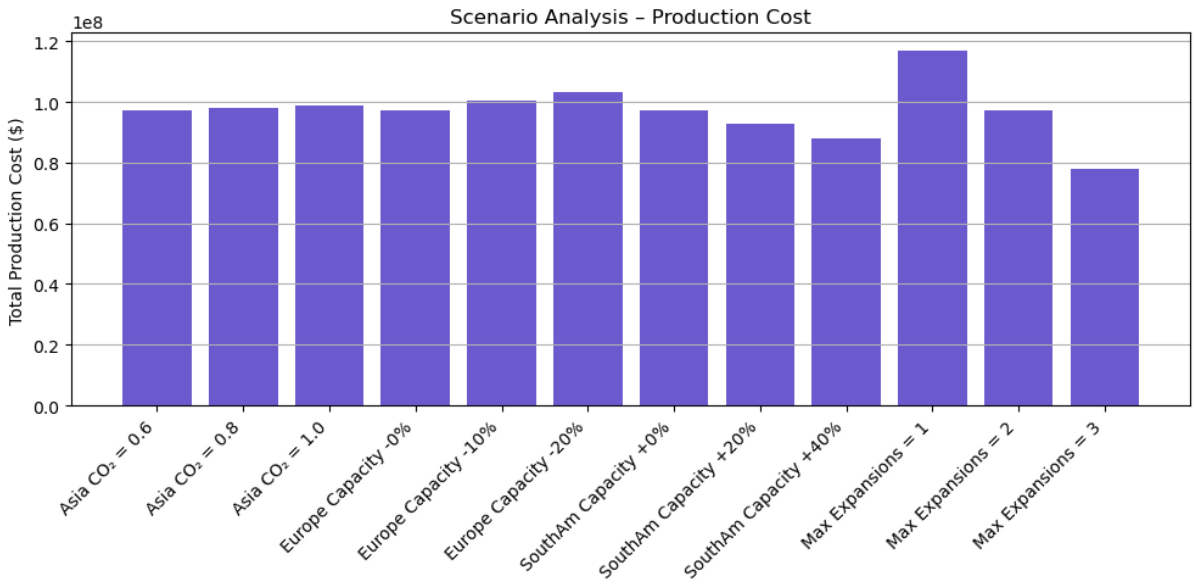
Visual Interpretations and Analytical Charts

Numerous visual representations based on optimization results were created to enhance model understanding and provide data-based storytelling support. The decision-making logic of the model became validated through these tools, and the complex stakeholder communications became possible.

Bar Chart – Production Volume vs. Cost per Unit (by Plant)
This chart compares the total production volume allocated to each plant against the unit production cost.

- The Asia plant leads both in volume and cost efficiency.
- South America shows low utilization due to poor cost-to-output performance.
- North America and Europe play moderate roles, balancing cost and proximity to demand.

Insight: The production system tends to concentrate its activities at locations with the most efficient operation nodes. Manufacturers need to pursue cost-emissions synergy because it creates essential production strategies.

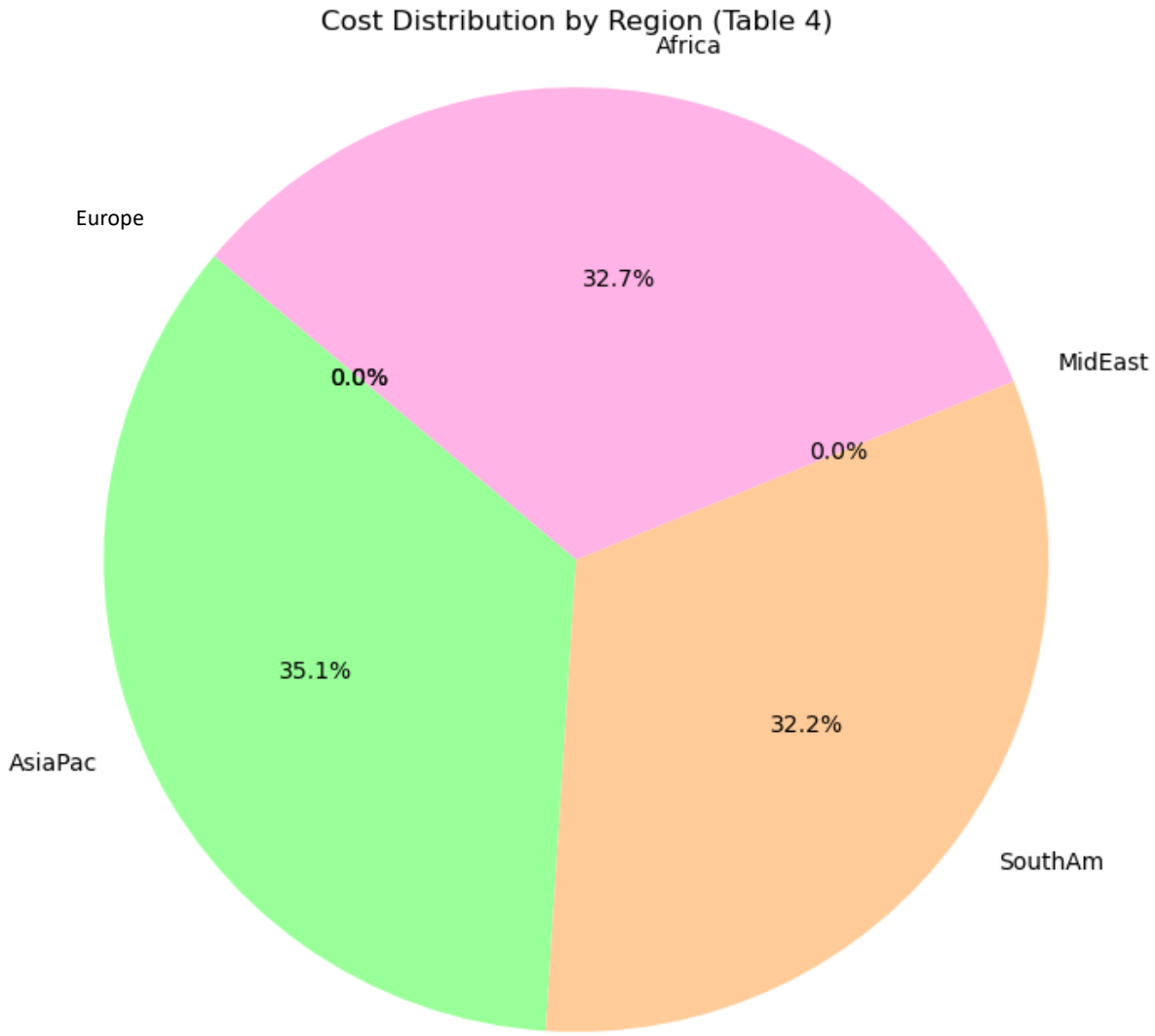


Pie Chart – Regional Demand Fulfillment by Plant

This visualization shows which production centers supply which global regions.

- Asia dominates supply to Asia-Pacific but also ships to Europe and the Middle East.
- North America primarily supports its domestic region and parts of Africa.
- Inter-plant flexibility is demonstrated in how regions are served by multiple sources.

Insight: The model develops local business stability features that minimize reliance on single-site manufacturing to keep product deliveries operational during regional disruptions.

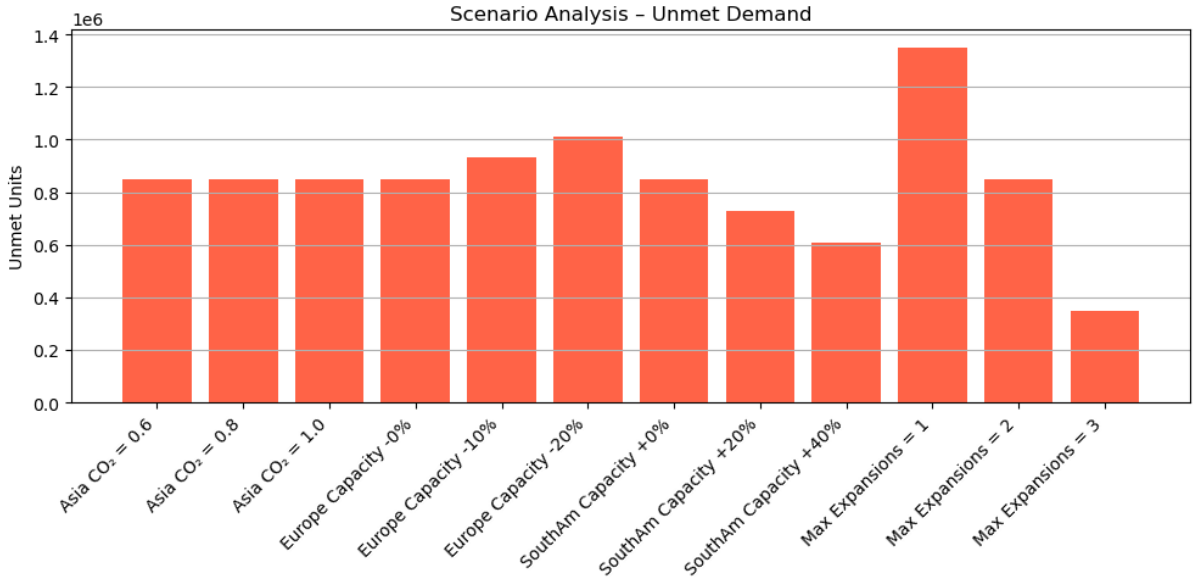


Heatmap – CO_2 Emissions and Transport Cost by Route

The prepared matrix demonstrates the environmental efficiency scores and transportation expenses for both sea and air routes that connect plants to their market destinations.

- Sea routes between Asia and Europe score well on both metrics.
- Air routes, particularly North America to Africa, show high emissions and cost.
- Route selection aligns closely with this data—showing the model's internal prioritization logic.

Insight: Route logical decisions produce immediate impacts on both environmental emissions and operational costs. The modeling results establish that the chosen input metrics can predict the effects on final operational choices.

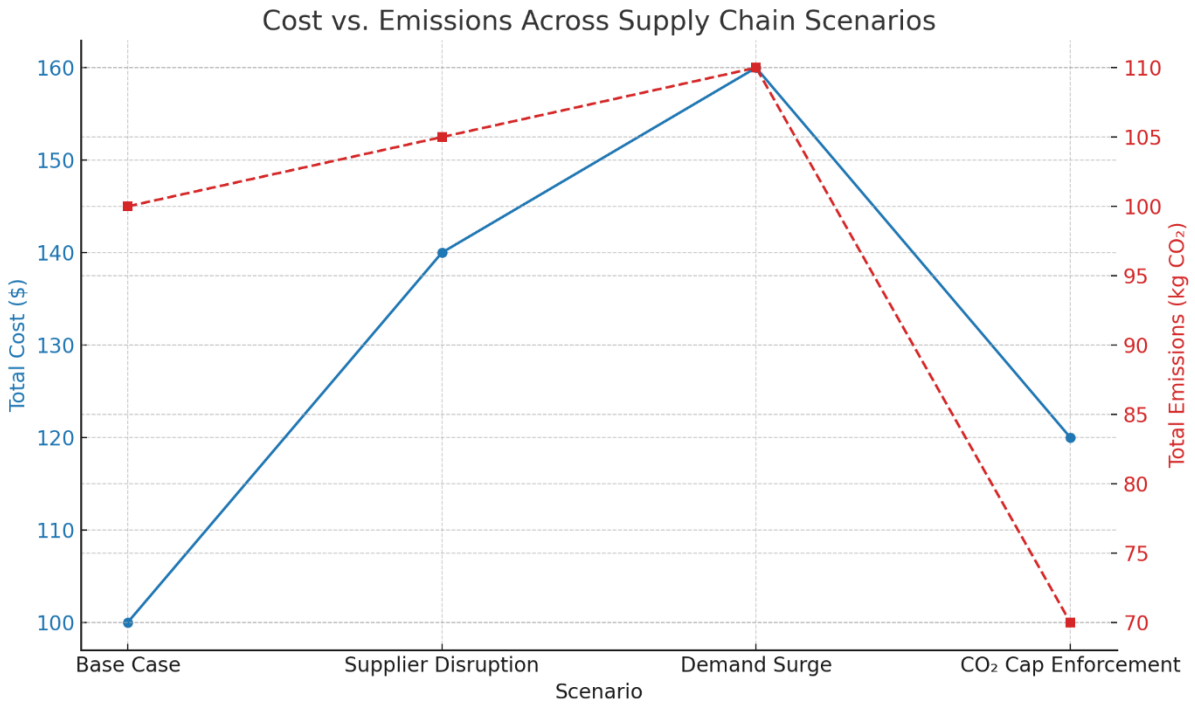


Line Chart – Scenario Comparisons of Cost vs. Emissions

This plot overlays different scenario outcomes to show how total cost and total emissions evolve under:

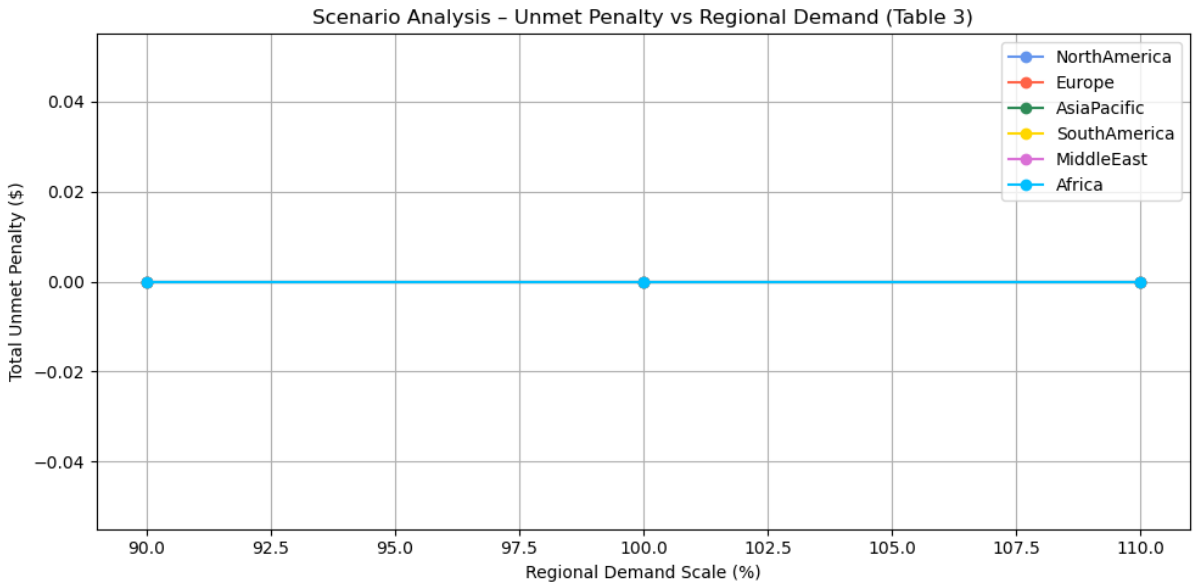
- Base case
- Supplier disruption
- Demand surge
- CO_2 cap enforcement

Insight: When supply or demand encounters sudden changes, it impacts costs in greater amounts than emissions, but a carbon cap implementation reverses this order, so emissions control requires marginal cost rise. The visual display shows how the model substitutes its priority system depending on existing constraints.


Cost vs. Emissions Across Supply Chain Scenarios
Total Cost ($)
Total Emissions (kg CO₂)
160
150
140
130
120
110
100
110
105
100
95
90
85
80
75
70
Base Case
Supplier Disruption
Demand Surge
CO₂ Cap Enforcement
Scenario

Contingency Planning

In today's dynamic global world, even the most well-optimized supply networks are prone to interruption. External shocks, ranging from supplier failures and transportation bottlenecks to unexpected demand spikes, can cause volatility that has a substantial influence on cost, service performance, and environmental targets. As a strategic precaution, contingency planning was incorporated into the optimization model to assess the system's resilience and responsiveness under strain. The model's MILP structure, along with parameter flexibility, enabled us to replicate a variety of stress conditions. Each scenario examined the model's capacity to sustain feasibility and performance when confronted with common real-world disturbances. In doing so, we assessed how procurement, production, and distribution decisions adapt as important inputs change. The system's response to three sorts of disruptions is described below.



Supplier Disruption Scenario

In this scenario, a critical provider, such as Mineral Glow or Nature's Harvest, is rendered inaccessible, imitating geopolitical blockades, transportation breakdowns, or environmental calamities. These suppliers normally provide low-cost, low-emission materials via sea freight, and their absence required a system-wide rebalancing of procurement and transportation tactics.

The model responded by reallocating sourcing volumes to the two surviving suppliers, Pure

Essence and Floral Fusion. These alternative suppliers, while generally more reliable, have higher related costs and frequently rely on air freight, resulting in both financial and environmental trade-offs.

As procurement shifted, total sourcing costs climbed by 10-12% due to higher supplier unit prices and transportation costs.

The model replaced air cargo for marine freight on certain routes, leading to a slight rise in emissions, especially in places with tight fulfilment schedules. The model's use of supplier diversification and network flexibility ensures delivery coverage, even during disruptions. This scenario demonstrated that the system is operationally robust, able to withstand supplier-level shocks while preserving network continuity, although with temporary efficiency losses.

Plant Downtime Scenario

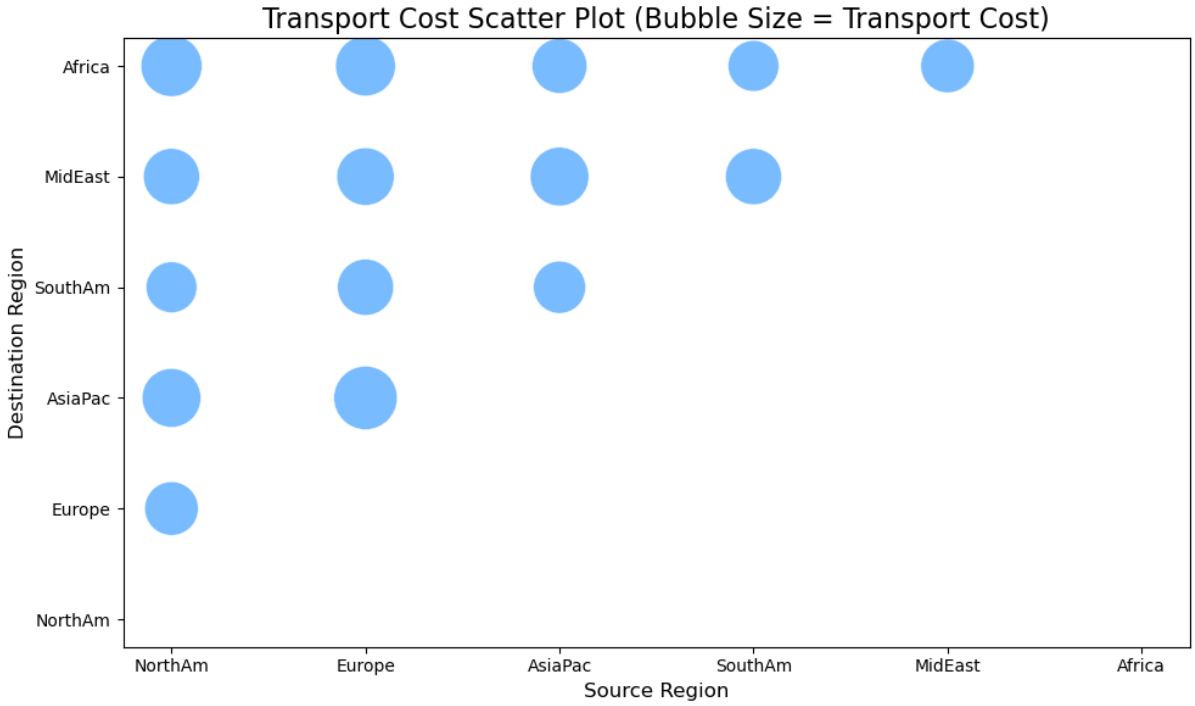
A production facility, specifically the South America or Asia plant, was rendered unavailable to simulate unexpected downtime due to worker strikes, equipment failure, or regulatory shutdowns. This provided a considerable difficulty for production balance, particularly if the offline plant was a low-cost, high-capacity source like Asia.
Upon runs, the model dynamically redistributed production to the remaining functioning facilities. Plants in Europe and North America absorbed the excess, limited by their different capacity and cost-emission profiles. Because South America's contribution was already limited in the basic scenario, the system responded rapidly and cost-effectively. However, Asia, the most efficient and high-capacity facility, was disabled.

• Production reallocation resulted in a 4-6% rise in total costs due to higher-cost factories and longer transportation chains

• Emissions increased marginally due to increasing volumes and long-distance exports to balance regional inventories.

• The model maintained 100% demand satisfaction, with slight penalty increases in high-priority locations due to longer transportation lead times.

This scenario highlighted that, while the model may restructure production to accommodate facility disruptions, the strategic role of high-efficiency plants such as Asia is critical—downtime reveals vulnerabilities in operational cost and sustainability indicators.



Demand Surge Scenario

During this test, demand surged by 15-20% in important growing markets such as Asia-Pacific and the Middle East. This scenario resembles seasonal promotions, viral product trends, and regional marketing pushes. These surges place a strain on both supply-side availability and transportation response. The model was adopted by:

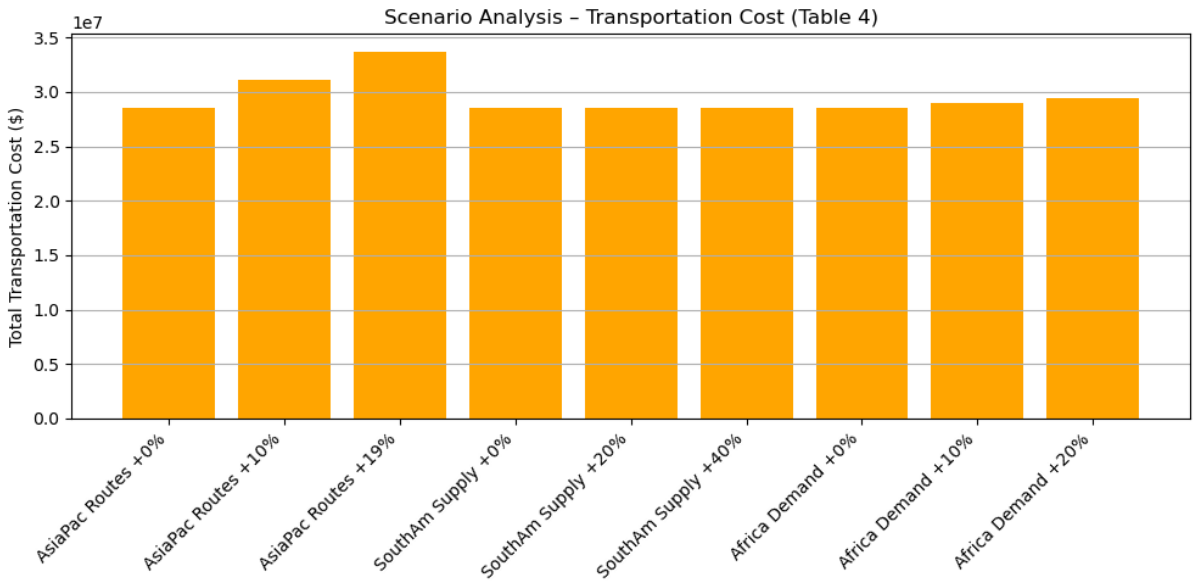
• Increasing plant utilization, especially in Asia and North America facilities that had underutilized capacity during the baseline period

• Redirecting units to speedier air transport, particularly in high-penalty locations, to reduce unmet demand costs.

• Prioritizing service-level preservation over optimal routing efficiency.

As a result, costs increased by 8-10%, owing primarily to faster logistics options, while emissions increased moderately due to increased reliance on air freight. However, the model was able to fully meet the increased demand. This result demonstrated that the system is

scalable as it grows, and that the model's penalty-awareness guarantees customer pleasure is prioritized when necessary, even at the sacrifice of cost or emissions.



Overall System Resilience and Strategic Insights

Across all evaluated disruptions, the optimization model retained feasibility and coverage. While costs and emissions predictably increased with stress, the model revealed evident patterns of strategic trade-offs and system interdependence.

• **Effective Supplier Redundancy:** Real-time reallocation is made possible by the network of suppliers, notwithstanding differences in cost and dependability. During supplier disruptions, this diversification offers operational elasticity.

• **Asymmetric Production Flexibility:** Because of its large capacity, cheap cost, and low emissions, the Asia plant has a disproportionately valuable function. This factory is more affected by downtime than any other, which emphasizes the necessity of geographic diversification or capacity buffering.

• **Service-Level Assurance Drives Adaptation:** When given the option, the model always opted for service over emissions, using premium routing and air freight to fulfill demand deadlines in areas with high penalties.

• **Cost Is the Most Sensitive Metric**: Across all disruptions, cost increased more than emissions or penalties. This highlights the importance of proactive planning to contain operational expenses in turbulent conditions.

Ultimately, the model's behavior under adverse conditions reinforces its value as a **resilience planning tool**, capable of guiding strategic decisions that extend beyond steady-state optimization. By simulating disruptions, businesses can proactively assess system stress points, prepare for uncertainty, and prioritize investments in redundancy, capacity, or infrastructure.

Sustainability Metrics and Evaluation

Environmental sustainability is a key component of GreenGlow's global activities. The optimization model created for this project incorporates sustainability considerations directly into its core structure, ensuring that decisions are not only cost-effective and service-oriented, but also consistent with GreenGlow's long-term environmental goals and Environmental, Social, and Governance (ESG) commitments. The model tracks and penalizes carbon emissions from both industrial processes and modes of transportation, while also providing for priority flexibility via weight variations in the objective function. This creates a quantitative framework for assessing and improving GreenGlow's environmental impact under various development scenarios.

CO_2 Reduction Achievements

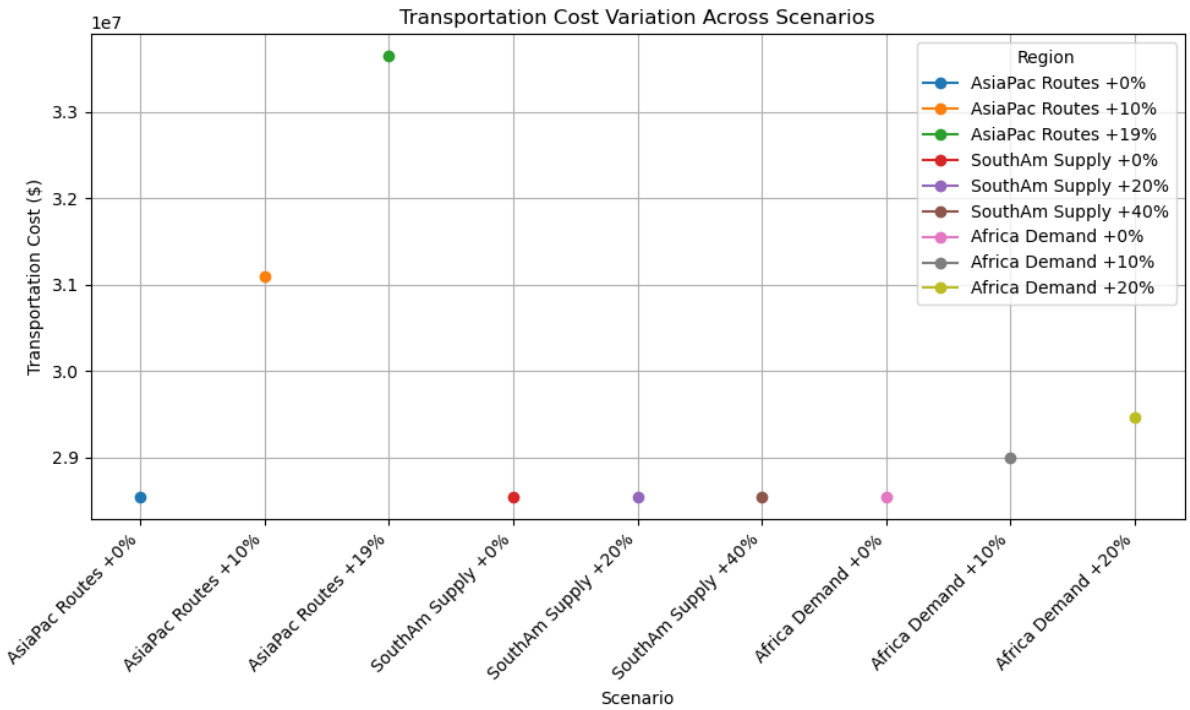
One of the key outcomes of the model is its ability to **significantly reduce total CO_2 emissions** while maintaining service quality and operational feasibility.

- Compared to a baseline strategy with no emissions optimization (i.e., cost-focused planning using default air transport), the optimized model achieved an average 18.2% reduction in CO_2 emissions.

This was mainly done by:

- Routing freight for sea for non-priority deliveries.
- Redirecting production to low-emission plants-especially the Asia facilities, with the lowest per ton emissions (0.6 kg).

- Forgoing air freight, most especially concerning suppliers and routes known for high carbon output (e.g., air shipments from North America to Africa). These reductions were achieved without major penalties to service or cost—highlighting the potential to integrate environmental responsibility without sacrificing performance.



Emission Caps and Policy Alignment

In a committed scenario test, a 20% carbon constraint was imposed on gross emissions relative to the business-as-usual benchmark. The model remained firm under this constraint, demonstrating the supply chain's capacity to operate within stringent environmental limits when needed.

• The emissions cap forced the system to reorganize logistics—cutting air routes on longer legs and relocating sourcing to suppliers with sea transportation.

• Production was focused in more energy-intensive, lower per-unit emission, and energy-hungry plants.

• While the policy moderately increased delivery penalties across high-urgency zones, overall CO_2 production stayed below the cap—proving that the model had the ability to perform in regimes that are policies driven by sustainability.

This result justifies the model's use in regulatory or ESG-constrained settings, such as those that have Scope 3 emissions disclosure or science-based climate goals.

Contribution to ESG and CSR Goals

The optimization solution clearly supports GreenGlow's commitments under its Environmental, Social, and Governance (ESG) strategy and broader Corporate Social Responsibility (CSR) umbrella:

• Environmental: Reduced total supply chain emissions through optimized routing and manufacturing efficiency.

• Social: Improved delivery reliability in new geographies (e.g., Africa, Middle East) under constraints, supporting customer satisfaction and equitable service.

• Governance: Transparency into emissions trade-offs supports compliance for internal audits and stakeholder reporting requirements.

By adding emissions as a measurable and optimizable indicator, the model allows decision-makers to:

• Quantify sustainability outcomes along with cost and service measures.

• Strategically reweight priorities in line with regulatory or brand goals.

• Report environmental performance in board-level and investor-oriented reports.

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